Relational Rationales from LLMs for Contextual Semantic Matching

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***Abstract*—Contextual semantic matching requires determining whether lexical units convey the same or related meanings based on their contextual usage. However, transformer-based models primarily encode the semantic evidence underlying such comparisons implicitly, leaving the value of explicit relational information underexplored. We investigate whether LLM-generated relational rationales can provide useful and complementary semantic information for contextual semantic matching. We propose a rationale-guided framework that generates label-agnostic rationales, applies leakage-aware filtering, and independently encodes rationales and contextual inputs before feature-level fusion. We evaluate the framework on the Word-in-Context (WiC) and Phrase Similarity (PiC-PS) tasks. Rationale-only representations outperform context-only representations on both tasks, while fusion improves over rationale-only representations by 2.30 percentage points on PiC-PS, reaching 81.60% accuracy. On WiC, fusion provides no additional improvement. Rationale shuffling substantially degrades performance on both tasks, indicating that the contribution of rationales depends on their coherent semantic content. These results show that LLM-generated relational rationales provide useful explicit semantic information, but their complementarity with contextual representations is task-dependent.**

***Index Terms*—Contextual Semantic Matching, Large Language Models (LLMs), Relational Rationale, Word-in-Context, Phrase Similarity, Representation Fusion**

I. Introduction

Contextual semantic matching determines whether lexical units convey the same or related meanings based on their usage in context. It is fundamental to language understanding and supports applications such as information retrieval [1, 2], machine translation [3, 4], question answering [5], and text summarization [6]. The task is challenging because the same lexical form can express different meanings across contexts, while different expressions can convey similar meanings depending on their usage.

The Word-in-Context (WiC) [7] and Phrase-in-Context (PiC) [8] benchmarks evaluate contextual semantic matching at the word and phrase levels, respectively. WiC determines whether a target word has the same meaning in two contexts, while PiC Phrase Similarity (PiC-PS) determines whether two target phrases convey similar meanings within a shared context. Transformer-based models achieve strong performance on these tasks through contextualized representations, but the semantic evidence captured by these representations remains implicit. This motivates investigating whether explicit relational semantic information can provide additional evidence for contextual matching.

Large language models (LLMs) can express such information through natural-language descriptions and rationales. Prior work has explored LLM-generated semantic descriptions for WiC [9] and rationale-augmented prompting for contextual word-meaning tasks [10], while Meng et al. [11] investigated instruction-following and chain-of-thought reasoning for phrase semantics on PiC-PS. These studies show that generated semantic explanations can provide useful information for contextual semantic matching. However, they primarily use rationales as standalone inputs, in-context reasoning signals, or components of prediction-level ensembling, rather than explicitly modeling their relationship with the original contextual representation.

A rationale may therefore provide useful semantic information without being complementary to the evidence captured by a contextual encoder. Generated rationales may also contain label-associated lexical cues, allowing models to exploit shortcuts rather than semantic content. Prior work has highlighted this label leakage problem in free-text rationales [12, 13, 14]. We therefore control the information contained in generated rationales before evaluating their contribution as semantic representations.

In this work, we investigate whether LLM-generated relational rationales provide independently useful and complementary information for contextual semantic matching. We generate label-agnostic relational rationales and apply leakage-aware filtering before encoding them independently from the original contextual inputs. The rationale and contextual representations are then evaluated separately and jointly through feature-level fusion, allowing us to distinguish the utility of relational information from its complementarity with contextual representations.

Experiments on WiC and PiC-PS reveal a task-dependent pattern. Rationale-only representations outperform context-only representations on both tasks, indicating that the generated rationales contain useful semantic information. Fusion further improves performance on PiC-PS by 2.30 percentage points, reaching 81.60% accuracy, but provides no additional

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improvement on WiC. Rationale shuffling substantially degrades performance on both tasks, indicating that the contribution depends on the coherent semantic content of the rationales rather than their lexical presence alone. Instance-level analysis further shows that the two representations make complementary contributions on PiC-PS, whereas fusion closely follows the rationale representation on WiC. These results indicate that the usefulness of LLM-generated relational rationales does not necessarily imply complementarity with contextual representations, and that this relationship depends on the semantic matching task.

The main contributions of this work are as follows:

1) We investigate LLM-generated relational rationales as an explicit source of semantic information for contextual semantic matching, with rationale generation performed without access to gold task labels.
2) We develop a rationale-guided representation framework that applies leakage-aware filtering and independently learns rationale and contextual representations before feature-level fusion.
3) We systematically analyze rationale-only, context-only, and fused representations on WiC and PiC-PS, revealing task-dependent differences in the utility and complementarity of relational rationales.

II. Proposed Framework

Our rationale-guided semantic matching framework investigates whether LLM-generated relational semantics can complement contextual representations. It comprises four components: (1) relational rationale generation, (2) leakage-aware rationale filtering, (3) independent representation learning from rationales and contextual inputs, and (4) feature-level fusion for semantic classification. The overall architecture is illustrated in Fig. 1.

A. Relational Rationale Generation

For each instance, the target lexical unit is identified and marked in the corresponding contextual input. For WiC, the annotated 0-indexed target position is used to identify the intended occurrence of the target word. For PiC-PS, where target positions are unavailable, the target phrase is located through exact substring matching, prioritizing whole-word matches and using case-insensitive prefix matching as a fallback.

The marked inputs are provided to Gemini 2.5 Flash to generate a concise expert-style relational rationale. The prompt asks the model to describe the target usage in terms of its sense, domain, contextual cues, and how its meaning or function changes or remains consistent across the inputs. The model is instructed not to state or imply the gold semantic relation. Its explicit yes/no judgment is placed after a designated end token and discarded before downstream encoding; it is retained only for offline analysis.

B. Leakage-Aware Rationale Filtering

Generated rationales may contain lexical patterns correlated with task labels even when explicit label prediction is discouraged. To reduce this potential leakage, we perform data-driven filtering using the training split only. Following Monroe et al. [15], informative-Dirichlet-prior log-odds z-scores are computed to identify words disproportionately associated with either class.

Target words in WiC and target-phrase tokens in PiC-PS are excluded from masking. A word is masked when its absolute z-score exceeds 3.0 and its class-frequency ratio is at least 2.5. The resulting word lists are applied consistently across all data splits. This procedure reduces readily exploitable label-associated lexical cues while preserving the semantic content of the generated rationales.

C. Dual-Encoder Representation Learning

The filtered rationales and original contextual inputs are encoded using separate transformer models. The rationale branch uses DeBERTa-v3$_{\text{Large}}$ [16], while the context branch uses ELECTRA$_{\text{Large}}$ [17]. Each encoder receives only its corresponding input: the rationale encoder receives the filtered rationale, and the context encoder receives the marked contextual input.

The two encoders are fine-tuned independently using the corresponding gold task labels. Following fine-tuning, the final-layer `[CLS]` representation from each encoder is extracted as its feature vector. The encoders are then frozen for the fusion stage.

D. Feature Fusion and Classification

The independently learned rationale and contextual representations are combined through feature-level fusion. Let h_r and h_c denote the final-layer `[CLS]` representations from the rationale and context branches, respectively. They are concatenated to form the joint representation

$$h = [h_r; h_c]. \tag{1}$$

The joint representation is passed to a multilayer perceptron (MLP) for binary classification. Both encoder branches remain frozen during fusion training, and only the MLP classification head is optimized. This separates independent representation learning from the subsequent combination of relational and contextual features.

III. Experimental Setup

A. Datasets

We evaluate the proposed framework on two contextual semantic matching benchmarks at different lexical granularities: Word-in-Context (WiC) [7] at the word level and the Phrase Similarity (PiC-PS) subtask of the Phrase-in-Context benchmark [8] at the phrase level.

WiC. Given a target word and two sentences containing it, the task determines whether the word is used with the same sense in both contexts. We use the standard splits, consisting of 5428 training, 638 validation, and 1400 test instances.

PiC-PS. Each instance contains two target phrases within a shared context, and the task determines whether the phrases

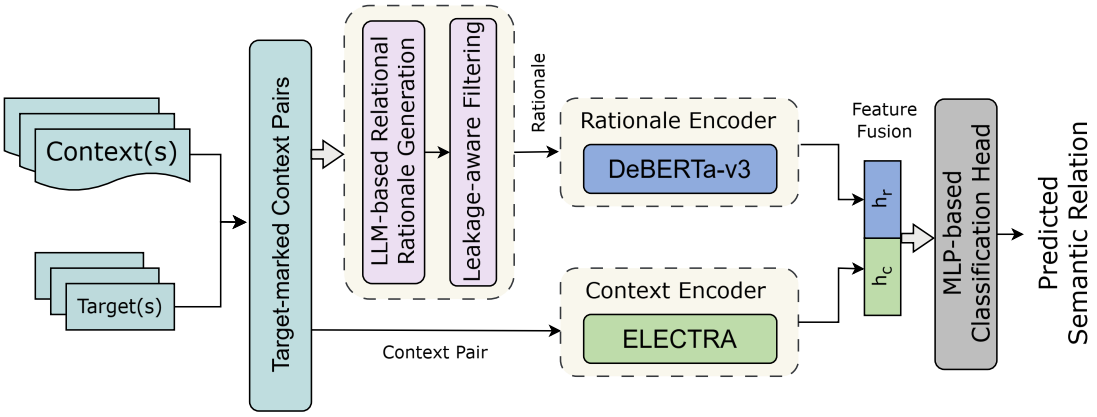


Fig. 1. Overview of the proposed rationale-guided semantic matching framework.

convey similar meanings in that context. We use 7004 training, 1000 validation, and 2000 test instances, consistent with prior work.

B. Baselines

We compare the proposed framework with three categories of previously reported systems: embedding-based methods, fine-tuned transformer models, and instruction-based LLMs. For PiC-PS, we use results reported by Meng et al. [8, 11], covering embedding, fine-tuned, and LLM-based systems with different prompting strategies. For WiC, we include representative fine-tuned and instruction-based results from prior work [18, 19, 20], together with reported human performance.

We additionally evaluate controlled single-encoder baselines using the same data and labels as the proposed framework. Each baseline receives the marked context, the filtered rationale, or both inputs concatenated into a single encoder. ELECTRA$_{\text{Large}}$ and DeBERTa-v3$_{\text{Large}}$ are evaluated under these settings to separate the contribution of rationale information from that of independent rationale and context encoding.

C. Implementation Details

All experiments are conducted using the HuggingFace `Trainer` [21] with mixed-precision (FP16) training on the Lightning AI platform [22], using a single NVIDIA T4 GPU and a fixed random seed of 42. Gemini 2.5 Flash [23] relational rationales are generated once for each instance using Google Cloud Batch Inference [1] with deterministic decoding and extended thinking disabled, and cached for subsequent experiments.

Rationales are filtered using the leakage-aware procedure described in Section II-B. Informative Dirichlet-prior log-odds z-scores [15] are computed on each training split, and words satisfying $|z| > 3.0$ and a class-frequency ratio of at least 2.5 are masked, excluding target words in WiC and target-phrase tokens in PiC-PS. Four words are masked for WiC and twelve for PiC-PS, retaining 99.6% and 99.7% of rationale words, respectively. A TF–IDF unigram logistic-regression probe with a maximum vocabulary of 5,000 terms and five-fold stratified cross-validation is used to assess residual lexical information. Filtering reduces probe accuracy from 69.5% to approximately 67% on WiC and from 62.7% to approximately 62% on PiC-PS.

For representation learning, DeBERTa-v3$_{\text{Large}}$ [16] and ELECTRA$_{\text{Large}}$ [17] are fine-tuned independently using AdamW [24], with weight decay of 0.01 and early stopping with patience 3. Maximum sequence lengths are 128 and 512 tokens for rationale and context inputs, respectively. The rationale and context branches use batch sizes of 16 and 8 with two and four gradient-accumulation steps, respectively. Learning rates are 1×10^{-5} and 3×10^{-6}, with linear warmup over 10% and 20% of the training steps, respectively, for up to 10 epochs.

Following encoder fine-tuning, the representations are frozen and concatenated for fusion. The resulting 2,048-dimensional representation is passed to an MLP with 512 hidden units, ReLU activation, and dropout of 0.1, followed by a linear classification layer. Only the fusion head is optimized at this stage, using a learning rate of 1×10^{-4}. Performance is reported using classification accuracy on the test sets.

D. Results and Analysis

1) Main Results: Table I compares the proposed framework with previously reported systems and controlled single-encoder baselines on WiC and PiC-PS. The controlled baselines use the same inputs as the proposed framework but encode them jointly in a single transformer.

The proposed framework achieves 75.21% accuracy on WiC and 81.60% on PiC-PS. On WiC, it improves over BERT-large and RoBERTa-large but remains below the strongest reported fine-tuned result and human performance. On PiC-PS, it exceeds the strongest previously reported fine-tuned result by 12.30 percentage points and the strongest instruction-based LLM result by 7.90 points.

The controlled baselines reveal a task-dependent effect. On PiC-PS, single-encoder variants reach approximately 80%, while the proposed dual-encoder fusion improves performance to 81.60%. On WiC, the single-encoder DeBERTa-v3$_{\text{Large}}$

[1] https://cloud.google.com/vertex-ai

TABLE I

MAIN PERFORMANCE COMPARISON, TEST ACCURACY (%). BEST SCORE IN EACH GROUP IS UNDERLINED; OVERALL BEST IS BOLDFACED. PIC-PS BASELINE SCORES ARE REPORTED BY MENG ET AL. [11]. WIC BASELINE SCORES ARE TAKEN FROM PRIOR REPORTED RESULTS [18, 19, 20].

Model	**WiC**	**PiC-PS**
Baselines (Embedding)		
BERT	–	64.1
SpanBERT	–	64.0
SpanBERT-Large	–	<u>66.3</u>
SentenceBERT	–	60.3
PhraseBERT	–	63.4
SimCSE	–	62.5
Baselines (Fine-tuned)		
BERT-large	69.6	68.9
RoBERTa-large	69.9	–
SpanBERT-Large (7K training ex.)	–	<u>69.3</u>
ERNIE 3.0	77.4	–
PaLM 540B	77.4	–
Vega v2	77.4	–
Turing NLR v5	77.1	–
DeBERTa / Turing NLR v4	77.5	–
T5+UKG	<u>77.9</u>	–
Baselines (LLM, instruction-based)		
GPT-3, few-shot (32-ex.)	49.4	–
GPT-3.5-Turbo, 0-shot	–	62.1
GPT-3.5-Turbo, 2-shot	–	70.8
GPT-4-Turbo, 0-shot	–	<u>73.7</u>
GPT-4-Turbo, 4-shot	–	73.5
GPT-4-Turbo, 4-shot + CoT	–	72.3
Human performance	80.0	–
Ablation (Single Encoder: filtered rationale + context)		
Single encoder ($\text{ELECTRA}_{\text{Large}}$)	75.21	<u>80.30</u>
Single encoder ($\text{DeBERTa-v3}_{\text{Large}}$)	<u>75.57</u>	80.25
Proposed Framework	75.21	**81.60**

result of 75.57% slightly exceeds the proposed 75.21%, indicating no benefit from dual-encoder fusion. We therefore analyze the individual representations and their interaction in the following subsection.

2) Analysis of Representation Contributions: Table II isolates the contribution of the contextual and rationale representations by evaluating each branch independently and comparing them with the fused representation.

TABLE II

REPRESENTATION-LEVEL COMPARISON, TEST ACCURACY (%). BEST CONFIGURATION PER TASK IS BOLDFACED.

Configuration	**WiC**	**PiC-PS**
Context-only ($\text{ELECTRA}_{\text{Large}}$)	72.57	78.50
Rationale-only ($\text{DeBERTa-v3}_{\text{Large}}$)	75.50	79.30
Fusion (Proposed Framework)	75.21	**81.60**

The rationale-only representation outperforms the context-only representation on both tasks, by 2.93 percentage points on WiC and 0.80 points on PiC-PS. This indicates that the generated rationales contain task-relevant semantic information even without access to the original contextual input during classification.

The effect of combining the two representations is task-dependent. On PiC-PS, fusion improves over the rationale-only branch by 2.30 points, reaching 81.60%, indicating complementary information between the contextual and relational representations. On WiC, fusion reaches 75.21%, slightly below the rationale-only result of 75.50%. Together with the single-encoder results in Table I, this suggests that the context branch provides limited additional signal once the rationale representation is available for word-sense classification.

These results show that the standalone usefulness of relational information does not necessarily imply complementarity with contextual representations. We next examine whether the rationale contribution depends on its coherent semantic content.

3) Rationale Shuffling Analysis: To determine whether rationale effectiveness depends on its instance-specific semantic content, we corrupt only the rationale input at test time by randomly permuting the words within each instance's own rationale. This preserves the original rationale vocabulary while disrupting its semantic organization. The context representation remains unchanged, and the experiment is repeated with three random permutation seeds.

TABLE III

RATIONALE SHUFFLING ANALYSIS, TEST ACCURACY (%). EACH SHUFFLED PERMUTATION USES A DIFFERENT RANDOM SEED TO PERMUTE THE WORDS WITHIN EACH INSTANCE'S OWN RATIONALE. CONTEXT-ONLY ACCURACY IS REPEATED AS A REFERENCE.

Configuration	**WiC**	**PiC-PS**
Correctly paired rationale	**75.21**	**81.60**
Shuffled, seed 42	49.43	58.95
Shuffled, seed 43	52.64	59.55
Shuffled, seed 44	52.36	60.45
Shuffled, mean (std.)	51.48 (1.45)	59.65 (0.62)
Gap (paired − shuffled mean)	23.74	21.95
Reference: context-only	72.57	78.50

Shuffling causes a substantial performance drop on both tasks, from 75.21% to a mean of 51.48% on WiC and from 81.60% to 59.65% on PiC-PS. Because the permutation is performed within each instance, the original rationale vocabulary is preserved while its coherent semantic content is disrupted. The resulting degradation indicates that the model relies on the structured semantic information provided by the correctly generated rationales rather than merely on their lexical presence.

Shuffled performance also falls below the corresponding context-only accuracy on both tasks. Thus, the fusion model does not simply ignore an uninformative rationale and rely on the intact context branch. Instead, the corrupted rationale interferes with the learned combination of the two representations. This exposes a limitation of the feature-level fusion strategy: although it benefits from coherent rationale representations, it is not robust to substantial corruption of the rationale branch at inference time.

4) Qualitative and Error Analysis: To examine the instance-level behavior of the three representations, we categorize test instances according to whether the context-only, rationale-only, and fusion predictions are correct.

TABLE IV
ERROR TAXONOMY: PER-INSTANCE AGREEMENT WITH THE GOLD LABEL.

Category	WiC	PiC-PS
All three correct	877	1335
All three wrong	208	185
Rationale corrects context	176	160
Context corrects rationale	0	122
Fusion retains rationale error	135	92
Fusion retains context error	0	70
Fusion error (both correct)	4	21
Fusion rescue (both wrong)	0	15
Total	1400	2000

PiC-PS shows stronger complementarity: the rationale and context branches correct 160 and 122 instances missed by the other branch, respectively, while fusion additionally rescues 15 instances missed by both. In contrast, WiC has no instances where context corrects a rationale error or fusion rescues an error made by both branches. Fusion agrees with rationale-only on 99.71% of WiC instances, compared with 77.5% with context-only; on PiC-PS, the corresponding rates are 88.6% and 85.6%. McNemar tests show that fusion differs significantly from context-only on both WiC ($p = 0.042$) and PiC-PS ($p = 3.1 \times 10^{-4}$), while its difference from rationale-only is significant only on PiC-PS ($p = 2.8 \times 10^{-3}$; WiC $p = 0.125$).

TABLE V
REPRESENTATIVE PIC-PS EXAMPLES ILLUSTRATING COMPLEMENTARY CONTRIBUTIONS FROM THE CONTEXT AND RATIONALE BRANCHES.

Case 1: Context misled
Context 1: "...the building houses Heritage Bank's various departments and **contact centre** and has a branch on the ground floor."
Context 2: "...various departments and **meeting place for nonresident parents and children** and has a branch on the ground floor."
Gold: *not similar* **Context:** *similar* (✗) **Rationale:** *not similar* (✓)
Fusion: *not similar* (✓)

Case 2: Rationale misled
Context 1: "Portugal has a stable wolf population of 200–300, which is afforded **full protection**."
Context 2: "...which is afforded **complete defense**."
Gold: *not similar* **Context:** *not similar* (✓) **Rationale:** *similar* (✗)
Fusion: *not similar* (✓)

Table V presents two representative PiC-PS cases. In the first, the context branch is misled while the rationale and fusion predictions are correct; in the second, the rationale branch is misled while the context and fusion predictions are correct.

On WiC, context can provide the correct decision even when the rationale branch fails. Table VI shows a representative case where *expression* occurs in different senses in "a puzzled **expression**" and "manner of **expression**." Context-only predicts different senses correctly, whereas rationale-only and fusion predict the same sense. This pattern occurs in 135 of 1400 test instances (9.6%).

TABLE VI
REPRESENTATIVE WIC EXAMPLE WHERE FUSION FOLLOWS THE INCORRECT RATIONALE PREDICTION.

Context 1: "They stared at the newcomer with a puzzled **expression**."
Context 2: "His manner of **expression** showed how much he cared."
Gold: *different senses* **Context:** *different senses* (✓) **Rationale:** *same sense* (✗) **Fusion:** *same sense* (✗)

Overall, the instance-level results reinforce the aggregate finding that relational rationales are useful on both tasks, but their complementarity with contextual representations is substantially stronger on PiC-PS than on WiC.

IV. CONCLUSION

We presented a rationale-guided framework for contextual semantic matching that combines LLM-generated relational rationales, leakage-aware filtering, and independently learned rationale and contextual representations. Experiments on WiC and PiC-PS show that rationale representations outperform context-only representations on both tasks, while fusion provides a 2.30 percentage-point improvement on PiC-PS but no additional gain on WiC. Rationale shuffling further indicates that these improvements depend on the coherent semantic content of the generated rationales rather than their lexical presence alone. More broadly, our findings suggest that relational rationales can provide useful explicit semantic information, but their value as a complementary representation depends on the nature of the matching task.

Our study is limited by a single training seed, the use of different encoders across branches, residual lexical signals after filtering (WiC probe accuracy of approximately 67%), and the possibility that rationales encode Gemini's implicit relational judgments. Future work will examine these issues through multi-seed experiments, same-encoder branches, and Gemini-prediction baselines, while exploring gated fusion, robust rationale representations, and multilingual and low-resource settings.

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